AISSAOUI Ismail

State Engineer in Artificial Intelligence & Data Science

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RESEARCH SUMMARY

State Engineer (ESI-SBA) specializing in **Physics-Informed Machine Learning (PINN)**, **Interpretability**, and **Robustness** for industrial Digital Twins. Expert in integrating physical laws into deep learning architectures (Mamba-SSM/xLSTM) and developing quantitative indicators for model reliability. Proven track record in accelerating physics-based simulations for large-scale energy systems (Sonatrach).

EDUCATION

National School of Computer Science (ESI-SBA, Algeria) 2021 – 2026
State Engineer's Degree & MSc in Artificial Intelligence

RESEARCH PROJECTS

Physics-Informed Digital Twin — Sonatrach 2026

Development of a generative surrogate for gas turbine monitoring.

- Implemented custom **PINN loss functions** to enforce thermodynamic laws.
- Developed a dual-track architecture using **Mamba-SSM** and **xLSTM**.
- Investigated **Uncertainty Quantification** and reliability metrics for OOD (Out-of-Distribution) detection in industrial surrogates.
- Optimized for resource-constrained edge devices with **0.04ms** latency.

Industrial Safety & CV Monitoring (YOLOv8) 2025

Computerized image processing system for safety equipment detection and tracking. *Key skill: Computerized Image Processing & Object Tracking.*

Recognizini — Real-Time Face Recognition 2025

High-speed vision system with active learning and incremental embeddings.

Geo-RAG — Semantic Knowledge Retrieval 2025

Knowledge extraction from multi-source industrial and geospatial data.

Career Connect — Structural GNN Intelligence 2024 – 2025

Relational modeling of complex systems via Graph Neural Networks.

TECHNICAL SKILLS

- **Physics-Informed AI**: PINN (Loss design), Koopman Theory, Physics-Aware Loss.
- **Computer Vision**: YOLOv8, OpenCV, Image Segmentation, Active Learning.
- **Sequence Modeling**: Mamba-SSM, xLSTM, Informer, TCN.
- **Engineering**: PyTorch, C++, Docker, MLflow, CUDA, Linux, Git.

LANGUAGES

Arabic (Native), English (Advanced - Professional Research Level), French (Fluent).